• **MALFUNCTION INDICATOR LAMP (MIL)—MIL ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTCS) P1380, P1381, OR P1383 IN MEMORY—NO DRIVEABILITY CONCERN S—VEHICLES WITH 2.0L ENGINE** • **ENGINE—2.0L—ENGINE OIL CONTAMINATION WITH DIAGNOSTIC TROUBLE CODES (DTCS) P1380, P1381, OR P1383 IN MEMORY—NO DRIVEABILITY CONCERNS**	**Article No. 00-3-7**

FORD: 1998-2000 CONTOUR, ESCORT ZX2

MERCURY: 1998-2000 MYSTIQUE
1999-2000 COUGAR

ISSUE

The Malfunction Indicator Lamp (MIL) may illuminate with Diagnostic Trouble Codes (DTCs) P1380, P1381 or P1383 stored in continuous memory on some vehicles. Driveability concerns may or may not be present. This may be caused by engine oil contamination.

ACTION

Refer to the following Service Procedure for diagnostic and repair procedures.

SERVICE PROCEDURE

The Variable Cam Timing (VCT) sprocket retards the exhaust camshaft up to 30 degrees producing exhaust valve overlap. This controlled overlap allows exhaust gases to enter the combustion chamber, effectively providing Exhaust Gas Recirculation (EGR) flow to the engine without the use of an EGR valve. The VCT holds the exhaust camshaft fully advanced by internal spring pressure in its normal, uncontrolled state. For example, if the VCT solenoid is unplugged. When retard is commanded by the Powertrain Control Module (PCM), engine oil pressure is directed through the VCT solenoid to overcome internal return spring force in the VCT sprocket.

Parameter Identification Displays (PIDs) Relevant To VCT:

CAMDCR (0 to 99%) is the on-time of the VCT solenoid as requested by the PCM.

CAMERR (0 to 30 Degrees) is how many degrees the exhaust cam is “out” from the requested position. Values “+” mean the cam is not retarded enough and values “-” indicate the cam is over retarded.

RCAM (-15 to +45 Degrees) represents actual position of the cam relative to the crankshaft.

Normal PID Values (Approximate):

Warm idle (exhaust cam fully advanced/no EGR effect): CAMDCR = 0%, CAMERR = 0 degrees, RCAM = -15 degrees.

Warm cruise: CAMDCR = 50%, CAMERR = 0 degrees, RCAM = 40 degrees.

DTC(s):

P1380 - VCT Solenoid Malfunction: Indicates a VCT circuit failure between the PCM and the VCT solenoid, or possible VCT solenoid failure. Consult the appropriate Electrical and Vacuum Troubleshooting Manual (EVTM) for circuit location and description.

P1381 - VCT Over-Advanced: Indicates the exhaust camshaft would not retard to the position requested by the PCM.

P1383 - VCT Over-Retarded: Indicates the exhaust camshaft would not advance (return) to the position requested by the PCM.

Diagnosing DTCs P1380, P1381, Or P1383:

P1380 and P1381 stored together: Possible open VCT solenoid or circuit between the PCM and VCT. These conditions would not allow the cam to retard as requested by the PCM.

P1381 or P1383: Must NOT be set together and ONLY in Continuous Memory for the following to be accurate: Possible VCT sprocket is sticking intermittently while driving. Exhaust camshaft is out of desired position as monitored by the PCM.

This condition could be caused by engine oil contamination. Consult TSB 99-13-8 for further diagnostics.

P1381: Single DTC set. Must be a hard fault. Monitor the following PIDs, checking for these readings: CAMDCR reads 99%, CAMERR reads a high "+" value while driving, and RCAM reads -15% at idle. This condition may indicate a sticking VCT sprocket which is retarding the exhaust camshaft. TSB 99-13-8 may remedy this concern. If concern persists after the TSB flushing procedure, replace the VCT sprocket. Refer to Section 303-01 in the appropriate Workshop Manual for VCT sprocket replacement. If RCAM is ABOVE -15 deg at idle, suspect base cam timing. Check and adjust the timing belt per Section 303-01 in the appropriate Workshop Manual.

P1383: Single DTC set/hard fault/rough running engine. If RCAM reads +45 at idle, VCT sprocket is stuck in the retarded position (very unlikely due to return spring force acting on VCT sprocket). VCT flush procedure from TSB 99-13-8 may be effective. If concern persists, replace VCT solenoid. Refer to Section 303-01 in the appropriate Workshop Manual for VCT solenoid replacement. If RCAM reads other than +45 at idle, suspect base cam timing. Check and adjust the timing belt per Section 303-01 in the appropriate Workshop Manual.

PART NUMBER	PART NAME
F8CZ-6L713-AA	Variable Camshaft Timing Solenoid
F8CZ-6256-AA	Variable Camshaft Timing Sprocket
F8CZ-6M282-AA	Sprocket Bolt

OTHER APPLICABLE ARTICLES: 99-13-8

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
000307A	Replace VCT Solenoid/Sprocket (Not To Be Used With Operations "B" Or "C")	1.9 Hrs.
000307B	Replace VCT Sprocket Only (Not To Be Used With Operations "A" Or "C")	1.9 Hrs.
000307C	Check And Adjust Base Timing (Not To Be Used With Operations "A" Or "B")	1.9 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
6256	69
6L713	69

OASIS CODES: 206000, 499000, 690000, 698298